Residual Diagnostics

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Introduction

Residual diagnostics are the post-fit checklist that exposes most assumption violations in regression. Four standard plots do nearly all the heavy lifting: residuals against fitted values, the Normal quantile-quantile plot, scale-location, and residuals-against-leverage. Reading these plots fluently — recognising a funnel as heteroscedasticity, a U-shape as missed non-linearity, heavy tails in the Q-Q plot as non-Normal errors — is one of the core skills of applied regression analysis.

Prerequisites

A working understanding of regression assumptions (linearity, homoscedasticity, Normal residuals, independence) and basic familiarity with quantile-quantile plots.

Theory

1. **Residuals-vs-fitted**: checks linearity and equal variance simultaneously. Random scatter around zero indicates the linear-predictor structure is adequate; a systematic trend suggests missed non-linearity; a funnel shape indicates heteroscedasticity.
2. **Normal Q-Q**: ordered standardised residuals plotted against theoretical Normal quantiles. Points on the reference line indicate Normality; heavy tails appear as S-shaped curves; skew shows as asymmetric departure.
3. **Scale-location** ($\sqrt{|r_i|}$ against fitted values): isolates the variance check. A horizontal line indicates homoscedasticity; an upward or downward trend indicates the residual variance changes with the mean.
4. **Residuals-vs-leverage**: identifies influential points. Cook's distance contours overlaid on the plot flag observations whose removal would substantially change the fit.

Studentised residuals (`rstudent()`) divide by an estimate of the residual SD that excludes observation i — preferable to ordinary standardised residuals when leverage is high.

Assumptions

Classical OLS assumptions; analogous diagnostics exist for GLMs (deviance and Pearson residuals) and mixed models (within- and between-cluster residuals).

R Implementation

```r
library(car); library(performance)

fit <- lm(mpg ~ wt + hp, data = mtcars)

# Built-in 4-panel diagnostic
par(mfrow = c(2, 2))
plot(fit)
par(mfrow = c(1, 1))
```

```r
# performance's modern alternative
check_model(fit)

# Individual residual types
resid(fit)              # raw
rstandard(fit)          # standardised
rstudent(fit)           # studentised
```

Output & Results

Four standard diagnostic plots from `plot(fit)`. `performance::check_model()` produces a richer multi-panel diagnostic with posterior predictive checks (for Bayesian fits), influential observations flagged, and explicit pass/fail-style annotations.

Interpretation

A reporting sentence: "Residual diagnostics for the linear model (mpg on weight and horsepower) showed: random scatter in residuals-vs-fitted (no missed non-linearity); approximately Normal residuals with mild heavy-tail behaviour at both extremes; constant variance across the fitted range; and three observations with moderate Cook's distance (D_i between 0.15 and 0.30) but none exceeding the conventional cutoff." Always describe what each diagnostic showed; reporting "diagnostics were satisfactory" without specifics is uninformative.

Practical Tips

- Always plot residuals; summary statistics do not catch the patterns that visual diagnostics reveal.
- Use studentised residuals (`rstudent`) for outlier checks; they correct for leverage and have a standard t reference distribution.
- Funnel-shaped residuals-vs-fitted indicate heteroscedasticity; remedies include log-transforming the response, weighted least squares, or robust HC3 standard errors.
- U-shape in residuals-vs-fitted indicates missed non-linearity; add polynomial terms, splines, or interactions.
- Heavy-tailed Q-Q plots motivate robust regression (`MASS::rlm`) or a heavier-tailed family (t regression, or robust ML in `lmrob`).
- Diagnostic plots are interpretive, not pass/fail; reasonable judgement is required and a small departure from ideal can be acceptable in large samples.

R Packages Used

Base R `plot.lm()` for the standard four-panel diagnostic; `car::residualPlots()` for component-plus-residual plots; `performance::check_model()` for an integrated tidy diagnostic; `DHARMa` for simulation-based residuals on GLMs and mixed models; `MASS::rlm()` for robust regression alternatives when residual diagnostics fail.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression
- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors
- One-way ANOVA and contrasts (emmeans)
- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures → mixed models
- GAMs with mgcv
- Non-linear regression with nls
- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI